

Практикум

C5:archive

Thirty-two questions, 137 marks. Every question the archive asks about vectors without a plane and without a vector product, from May 2021 to November 2025, in the order of the eight techniques rather than the order of the papers.

О практикуме

Тема разрезана по действию, а не по разметке. `geometry.vectors` и `geometry.vectors_3d` делят вопросы не по тому, что с векторами делают: прямые и их пересечение лежат в обеих, плоскости и векторное произведение — тоже в обеих. C5 берёт всё, что делается без плоскости и без векторного произведения: 96 баллов из 182 `geometry.vectors` (0.527) и 52 из 216 `geometry.vectors_3d` (0.241). 148 баллов по корпусу, 137 рабочих — ноябрь 2023 года стоит дважды, пирамида Q1 и прямые Q12(a). 58 баллов Paper 1, 71 Paper 2, 8 Paper 3.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-c5-vectors

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

C5 archive — vectors, lines and angles, all of it

Thirty-two questions, 137 marks.

Every question the archive asks about vectors without a plane and without a vector product, from May 2021 to November 2025, in the order of the eight techniques rather than the order of the papers.

No theory. No worked examples. The theory is in the practicum, `practicum-c5-vectors.ipynb`.

§	technique	questions	marks
1	Points and lengths	7	21
2	The scalar product	4	17
3	The angle between two vectors	4	14
4	The equation of a line	4	11
5	The angle between two lines	3	16
6	Where two lines meet	4	18
7	Parallel, meeting or skew	4	24
8	Motion along a line	2	16

The checks are the same ones the practicum uses and they store nothing: each is handed the points and lines exactly as the question prints them — `vec(1, -4, 0)`, `cartesian((x + 1)/2, y, 3 - z)`, `through(A, B)` — and a point or a letter the question asks for is handed **the conditions** that fix it, `parallelogram(A, B, C, D)`, `perpendicular(q, a)`, `meet(L1, L2)`. A line is accepted with any point on it and any parallel direction. Three significant figures, unless the question asks otherwise.

Parts of one question are split by technique. November 2025 TZ3 Q10 appears in §§ 4 and 7; each part repeats what it needs.

A *show that* is checked by the line before the given result: the scalar product before it is factorised, the cosine in terms of t , the parameter at the given point, the two parameters from two components.

Enter a line as `vec(point) + lam*vec(direction);` `lam` and `mu` are λ and μ .

Solutions are at the very bottom, deliberately far away.

```
import sys
sys.path.append('..')          # from
practicum/geometry to practicum/kit/
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
vec, line, cartesian, through, unknown

language('en')                 # this
notebook is in English, and so are the checks

print('ready; sympy', sp.__version__)
```

§ 1. Points and lengths

Seven questions, 21 marks. A vector between two points, a midpoint, a vertex, a length.

1.1 — May 2025 TZ3 Paper 1 Q11(a), (b), 4 marks

The points $A(1, -4, 0)$, $B(-3, -6, 2)$, $C(-1, -2, 4)$ and D form a parallelogram, $ABCD$, where D is diagonally opposite B .

(a) Find the coordinates of D .

The diagonals of the parallelogram, $[AC]$ and $[BD]$, intersect at point E .

(b) Find the coordinates of E .

```
A, B, C = vec(1, -4, 0), vec(-3, -6, 2),  
vec(-1, -2, 4)
```

```
D, E = unknown('D', 3), unknown('E', 3)
```

```
q1_1a = ...      # D
```

```
q1_1b = ...      # E
```

```
verify_find('1.1(a)', q1_1a, D,  
[parallelogram(A, B, C, D)])
```

```
verify_find('1.1(b)', q1_1b, E,  
[parallelogram(A, B, C, D), on(E, A, C),  
on(E, B, D)])
```

1.2 — November 2022 Paper 1 Q2(a), 4 marks

Consider a circle with a diameter AB , where A has coordinates $(1, 4, 0)$ and B has coordinates $(-3, 2, -4)$.

(a) Find **(i)** the coordinates of the centre of the circle; **(ii)** the radius of the circle.

```
A, B = vec(1, 4, 0), vec(-3, 2, -4)
centre, r = unknown('M', 3), unknown('r')

q1_2a = ...      # the centre
q1_2b = ...      # the radius

verify_find('1.2(a)(i)', q1_2a, centre,
[midpoint(centre, A, B)])
verify_find('1.2(a)(ii)', q1_2b, r, [Eq(2*r,
distance(A, B))])
```

1.3 — November 2023 TZ1 Paper 2 Q1(a), 2 marks

A pyramid has vertex V and rectangular base $OABC$. Point B has coordinates $(6, 8, 0)$ and point V has coordinates $(3, 4, 9)$.

(a) Find BV .

1.4 — May 2025 TZ2 Paper 2 Q2(a), 2 marks

A square-based right-pyramid has vertex $V(1, 7, 0)$. Point $X(-3, 4, 2)$ is the centre of the base $ABCD$.

(a) Find VX .

```
q1_3 = ...      # BV
q1_4 = ...      # VX

verify_find('1.3', q1_3, distance(vec(6, 8,
0), vec(3, 4, 9)))
verify_find('1.4', q1_4, distance(vec(1, 7,
0), vec(-3, 4, 2)))
```

1.5 — November 2022 Paper 2 Q12(a), (b), 3 marks

Consider the points $A(1, 2, 3)$, $B(k, -2, 1)$ and $C(5, 0, 2)$, where $k \in \mathbb{R}$.

(a) Write down $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

(b) Given that the points A , B and C lie on a straight line, show that $k = 9$.

For (b) enter the value of k that the line gives.

```
A, B, C = vec(1, 2, 3), vec(k, -2, 1), vec(5, 0, 2)
```

```
q1_5_AB = ...      # AB
```

```
q1_5_AC = ...      # AC
```

```
q1_5b = ...        # k
```

```
verify_find('1.5(a) AB', q1_5_AB, B - A)
```

```
verify_find('1.5(a) AC', q1_5_AC, C - A)
```

```
verify_find('1.5(b)', q1_5b, k, [parallel(B - A, C - A)])
```

1.6 — May 2022 TZ1 Paper 2 Q7(a), (b), 4 marks

Consider the vectors **a** and **b** such that

$$\mathbf{a} = \begin{pmatrix} 12 \\ -5 \end{pmatrix} \text{ and } |\mathbf{b}| = 15.$$

(a) Find the possible range of values for $|\mathbf{a} + \mathbf{b}|$.

Consider the vector **p** such that $\mathbf{p} = \mathbf{a} + \mathbf{b}$.

(b) Given that $|\mathbf{a} + \mathbf{b}|$ is a minimum, find **p**.

In (a) answer Interval(lowest, highest).

```

theta = symbols('theta')
a = vec(12, -5)
b = 15*vec(cos(theta), sin(theta))      #
every b with |b| = 15

q1_6a = ...      # Interval(lowest, highest)
q1_6b = ...      # p

verify_extent('1.6(a)', q1_6a, mag(a + b),
theta, Interval(0, 2*pi))
verify_optimum('1.6(b)', q1_6b, a + b, theta,
Interval(0, 2*pi), 'min')

```

1.7 — May 2023 TZ2 Paper 1 Q9(a), 2 marks

The diagram shows parallelogram $OABC$ with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$ and $|\mathbf{c}| = 2|\mathbf{a}|$, where $|\mathbf{a}| \neq 0$. The angle between $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is θ , where $0 < \theta < \pi$.

Point M is on $[AB]$ such that $\overrightarrow{AM} = k\overrightarrow{AB}$, where $0 \leq k \leq 1$ and $\overrightarrow{OM} \cdot \overrightarrow{MC} = 0$.

(a) Express $\overrightarrow{OM}$ and $\overrightarrow{MC}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.

```

la, theta = symbols('|a| theta',
positive=True)
a = la*vec(1, 0)
c = 2*la*vec(cos(theta), sin(theta))    # |c|
= 2|a|, the angle between them is theta
O = vec(0, 0)
B, M = unknown('B', 2), unknown('M', 2)
figure = [parallelogram(O, a, B, c,
names='OABC'), Eq(M - a, k*(B - a))]

q1_7_OM = ...    # OM in terms of a and c
q1_7_MC = ...    # MC in terms of a and c

verify_find('1.7 OM', q1_7_OM, M - O, figure)
verify_find('1.7 MC', q1_7_MC, c - M, figure)

```

§ 2. The scalar product

Four questions, 17 marks.

Perpendicular means zero; with letters, expand like brackets.

2.1 — May 2022 TZ1 Paper 2 Q7(c), 5 marks

Consider the vectors $\mathbf{a} = \begin{pmatrix} 12 \\ -5 \end{pmatrix}$ and $\mathbf{b}$ with $|\mathbf{b}| = 15$, and the vector $\mathbf{q} = \begin{pmatrix} x \\ y \end{pmatrix}$, where $x, y \in \mathbb{R}^+$.

(c) Find $\mathbf{q}$ such that $|\mathbf{q}| = |\mathbf{b}|$ and $\mathbf{q}$ is perpendicular to $\mathbf{a}$.

```
a = vec(12, -5)
q = unknown('q', 2)

q2_1 = ...      # q

verify_find('2.1', q2_1, q, [perpendicular(q,
a), length(q, 15), q[0] > 0, q[1] > 0])
```

2.2 — May 2023 TZ2 Paper 1 Q9(b), 3 marks

Parallelogram $OABC$ has $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$, $|\mathbf{c}| = 2|\mathbf{a}|$ and the angle θ between them.

M is on $[AB]$ with $\overrightarrow{AM} = k\overrightarrow{AB}$ and $\overrightarrow{OM} \cdot \overrightarrow{MC} = 0$.

(b) Hence, use a vector method to show that $|\mathbf{a}|^2(1 - 2k)(2 \cos \theta - (1 - 2k)) = 0$.

Enter $\overrightarrow{OM} \cdot \overrightarrow{MC}$ in terms of $|\mathbf{a}|$ (la), θ and k — the line before you factorise.

```
la, theta = symbols('|a| theta',
positive=True)
a = la*vec(1, 0)
c = 2*la*vec(cos(theta), sin(theta))
O = vec(0, 0)
B, M = unknown('B', 2), unknown('M', 2)
figure = [parallelogram(O, a, B, c,
names='OABC'), Eq(M - a, k*(B - a))]

q2_2 = ...          # OM · MC in terms of la,
theta and k

verify_find('2.2', q2_2, dot(M - O, c - M),
figure)
```

2.3 — May 2025 TZ3 Paper 2 Q6(a), 3 marks

Consider the vectors $\mathbf{a} = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -8 \\ 9 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} p \\ -6 \end{pmatrix}$, where $p \in \mathbb{R}$.

- (a)** Find an expression, in terms of p , for
(i) $\mathbf{a} \cdot \mathbf{c}$; **(ii)** $\mathbf{b} \cdot \mathbf{c}$.

```
p = symbols('p')
a, b, c = vec(-5, 7), vec(-8, 9), vec(p, -6)

q2_3_i = ...      # a · c
q2_3_ii = ...     # b · c

verify_find('2.3(i)', q2_3_i, dot(a, c))
verify_find('2.3(ii)', q2_3_ii, dot(b, c))
```

2.4 — November 2025 TZ1 Paper 3

Q1(f), 6 marks

The curve F has equation $y = \frac{k^2}{x}$ where $x \in \mathbb{R}$, $x \neq 0$ and $k \in \mathbb{R}$, $k \neq 0$.

The point $A \left(kt, \frac{k}{t} \right)$, where $t \in \mathbb{R}$, $t \neq \pm 1$, lies on F . The line normal to F at A intersects F again at point B ; in part (e) (ii) this gives $B \left(-\frac{k}{t^3}, -kt^3 \right)$.

From A , the line passing through the origin O intersects F again at point C . Points A , B and C form triangle ABC .

- (f)** Prove that $\hat{BCA}$ is a right angle.

Enter C and the vectors $\overrightarrow{CA}$ and $\overrightarrow{CB}$ in terms of k and t . The proof is that their scalar product simplifies to zero.

```
k, t = symbols('k t', positive=True)
A, B = vec(k*t, k/t), vec(-k/t**3, -k*t**3)
C = unknown('C', 2)
curve_and_line = [on(C, vec(0, 0), A),
Eq(C[0]*C[1], k**2), Ne(C, A)]

q2_4_C = ...      # C
q2_4_CA = ...     # CA
q2_4_CB = ...     # CB

verify_find('2.4 C', q2_4_C, C,
curve_and_line)
verify_find('2.4 CA', q2_4_CA, A - C,
curve_and_line)
verify_find('2.4 CB', q2_4_CB, B - C,
curve_and_line)
```

§ 3. The angle between two vectors

Four questions, 14 marks. $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$
, from the vertex.

3.1 — November 2023 TZ1 Paper 2 Q1(b), 4 marks

A pyramid has vertex V and rectangular base $OABC$. Point B has coordinates $(6, 8, 0)$, point C has coordinates $(6, 0, 0)$ and point V has coordinates $(3, 4, 9)$.

(b) Find the size of $\hat{BVC}$.

```
q3_1 = ...          # the angle BVC

verify_angle('3.1', q3_1, vec(6, 8, 0),
vec(3, 4, 9), vec(6, 0, 0))
```

3.2 — November 2022 Paper 2 Q7(a), 3 marks

Consider the vectors $\mathbf{u} = \mathbf{i} + \mathbf{j}$ and $\mathbf{v} = (\cos \frac{1}{n})\mathbf{i} + (\sin \frac{1}{n})\mathbf{j}$, where $n \in \mathbb{Z}^+$. Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$.

(a) Find an expression for $\cos \theta$ in terms of n .

```
n = symbols('n', positive=True, integer=True)
q3_2 = ...          # cos(theta) in terms of n

verify_angle('3.2', q3_2, vec(1, 1),
vec(cos(1/n), sin(1/n)), cosine=True)
```

3.3 — May 2025 TZ2 Paper 3 Q2(b)(i), 2 marks

The Earth is modelled as a sphere with its centre at the origin O . The North Pole

P has position vector $\mathbf{p} = \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix}$ and

Nairobi N has position vector $\mathbf{n} = \begin{pmatrix} 0 \\ 6 \\ 0 \end{pmatrix}$.

(b) (i) Use the scalar product to find the angle between $\mathbf{p}$ and $\mathbf{n}$.

3.4 — May 2025 TZ3 Paper 2 Q6(b), 5 marks

Consider the vectors $\mathbf{a} = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -8 \\ 9 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} p \\ -6 \end{pmatrix}$, where $p \in \mathbb{R}$. The angle between $\mathbf{a}$ and $\mathbf{c}$ is equal to the angle between $\mathbf{b}$ and $\mathbf{c}$.

(b) Find the value of p .

```
p = symbols('p')
a, b, c = vec(-5, 7), vec(-8, 9), vec(p, -6)

q3_3 = ...      # the angle between p and n
q3_4 = ...      # p

verify_angle('3.3', q3_3, vec(0, 0, 6),
vec(0, 6, 0))
verify_find('3.4', q3_4, p, [Eq(angle(a, c),
angle(b, c))])
```

§ 4. The equation of a line

Four questions, 11 marks. A point on it, a direction along it — and " $\mathbf{r} =$ ".

4.1 — May 2021 TZ1 Paper 1 Q11(a), 4 marks

Consider the line L_1 defined by the Cartesian equation $\frac{x+1}{2} = y = 3-z$.

(a) (i) Show that the point $(-1, 0, 3)$ lies on L_1 . **(ii)** Find a vector equation of L_1 .

4.2 — May 2025 TZ2 Paper 1 Q2(a), 2 marks

The line L_1 is defined by the Cartesian equation $\frac{x-1}{2} = \frac{y+2}{3} = z$.

(a) Find a vector equation of L_1 .

```

q4_1 = ...      # a vector equation of  $(x + 1)/2 = y = 3 - z$ 
q4_2 = ...      # a vector equation of  $(x - 1)/2 = (y + 2)/3 = z$ 

verify_line('4.1', q4_1, cartesian((x + 1)/2,
y, 3 - z))
verify_line('4.2', q4_2, cartesian((x - 1)/2,
(y + 2)/3, z))

```

4.3 — November 2025 TZ3 Paper 1 Q10(a), (b), 3 marks

The point $P(-1, 1, -13)$ lies on the line L_1 .

The line L_1 has a direction vector $\begin{pmatrix} 7 \\ 1 \\ 2 \end{pmatrix}$.

(a) Write down a vector equation for L_1 in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$.

(b) Find a vector equation for line L_2 in the form $\mathbf{s} = \mathbf{c} + \mu\mathbf{d}$, given that L_2 passes through the points $A(2, -4, 2)$ and $B(7, -6, 1)$.

```

q4_3a = ...      # L1: vec(...) +
lam*vec(...)
q4_3b = ...      # L2: vec(...) + mu*vec(...)

verify_line('4.3(a)', q4_3a, line(vec(-1, 1,
-13), vec(7, 1, 2)))
verify_line('4.3(b)', q4_3b, through(vec(2,
-4, 2), vec(7, -6, 1)))

```

4.4 — November 2025 TZ1 Paper 2 Q12(f), 2 marks

The line L_3 is normal to the plane Π and passes through the point of intersection of L_1 and L_2 . In part (e) its equation is

$$\mathbf{r}_3 = \begin{pmatrix} 1 \\ 8 \\ -2 \end{pmatrix} + \gamma \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}.$$

(f) Given that the point $S(-11, 5, 7)$ lies on the line L_3 , find γ .

```

gamma = symbols('gamma')
L3 = line(vec(1, 8, -2) + gamma*vec(-4, -1,
3))

q4_4 = ...      # gamma

verify_line_parameter('4.4', q4_4, L3,
vec(-11, 5, 7))

```

§ 5. The angle between two lines

Three questions, 16 marks. The directions only, and the acute angle.

5.1 — May 2021 TZ1 Paper 1 Q11(b), 8 marks

Consider the line L_1 defined by $\frac{x+1}{2} = y = 3 - z$ and a second line L_2 defined by $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + t \begin{pmatrix} a \\ 1 \\ -1 \end{pmatrix}$, where $t \in \mathbb{R}$ and $a \in \mathbb{R}$.

(b) Find the possible values of a when the acute angle between L_1 and L_2 is 45° .

Paper 1: exact values, all of them.

```
a = symbols('a')
L1 = cartesian((x + 1)/2, y, 3 - z)
L2 = line(vec(0, 1, 2) + t*vec(a, 1, -1), t)

q5_1 = [...]      # all possible values of a

verify_find('5.1', q5_1, a, [Eq(angle(L1,
L2), pi/4)], exact=True)
```

**5.2 — May 2022 TZ2 Paper 2 Q11(c),
4 marks**

Two airplanes, A and B , have position vectors with respect to an origin O given respectively by

$$\mathbf{r}_A = \begin{pmatrix} 19 \\ -1 \\ 1 \end{pmatrix} + t \begin{pmatrix} -6 \\ 2 \\ 4 \end{pmatrix}$$

$$\mathbf{r}_B = \begin{pmatrix} 1 \\ 0 \\ 12 \end{pmatrix} + t \begin{pmatrix} 4 \\ 2 \\ -2 \end{pmatrix}$$

where t represents the time in minutes and $0 \leq t \leq 2.5$. Entries in each column vector give the displacement east of O , the displacement north of O and the distance above sea level, all measured in kilometres.

(c) Find the acute angle between the two airplanes' lines of flight. Give your answer in degrees.

```

rA = vec(19, -1, 1) + t*vec(-6, 2, 4)
rB = vec(1, 0, 12) + t*vec(4, 2, -2)

q5_2 = ...          # the acute angle, degrees

verify_angle('5.2', q5_2, line(rA, t),
line(rB, t), deg=True)

```

5.3 — May 2023 TZ1 Paper 1 Q12(a), 4 marks

Two lines, L_1 and L_2 , intersect at point P . Point $A(2t, 8, 3)$, where $t > 0$, lies on L_2 .

The acute angle between the two lines is $\frac{\pi}{3}$.

The direction vector of L_1 is $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, and

$$\overrightarrow{PA} = \begin{pmatrix} 2t \\ 0 \\ 3+t \end{pmatrix}.$$

(a) Show that $4t = \sqrt{10t^2 + 12t + 18}$.

Enter $\cos \theta$ in terms of t , before you set it equal to $\frac{1}{2}$.

```
t = symbols('t', positive=True)

q5_3 = ...          # cos of the angle between
the lines, in terms of t

verify_angle('5.3', q5_3, line(vec(0, 0, 0),
vec(1, 1, 0)), line(vec(0, 0, 0), vec(2*t, 0,
3 + t)), cosine=True)
```

§ 6. Where two lines meet

Four questions, 18 marks. Two parameters, two equations, the third checked.

6.1 — May 2025 TZ2 Paper 1 Q2(b), 3 marks

The line L_1 is defined by the Cartesian equation $\frac{x-1}{2} = \frac{y+2}{3} = z$. A second line L_2 is defined by the vector equation $\mathbf{r} = \begin{pmatrix} 0 \\ 4 \\ -8 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$, where $t \in \mathbb{R}$.

(b) Find the coordinates of the point where L_1 and L_2 intersect.

```
q6_1 = ...          # the point of intersection

verify_meet('6.1', q6_1, cartesian((x - 1)/2,
(y + 2)/3, z), line(vec(0, 4, -8) + t*vec(1,
0, 2)))
```

6.2 — November 2025 TZ1 Paper 2

Q12(a), 3 marks

The equations of two lines, L_1 and L_2 , are given by

$$\begin{aligned} L_1 : \mathbf{r}_1 &= \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} + s \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \text{ where } s \in \mathbb{R}, & L_2 : \frac{x+1}{2} \\ &= y - 7 \\ &= \frac{z+5}{3} \end{aligned}$$

(a) Show that the position vector of the point of intersection of L_1 and L_2 is $\begin{pmatrix} 1 \\ 8 \\ -2 \end{pmatrix}$.

Enter the value of s at the point of intersection.

```
s = symbols('s')

q6_2 = ...      # s at the point of
intersection

verify_line_parameter('6.2', q6_2,
line(vec(5, 4, 2) + s*vec(1, -1, 1), s),
vec(1, 8, -2))
```

6.3 — November 2023 TZ1 Paper 2

Q12(a), 5 marks

Line L is given by the vector equation

$$\mathbf{r}_1 = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + s \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} \text{ where } s \in \mathbb{R}.$$

Line M is given by the vector equation

$$\mathbf{r}_2 = \begin{pmatrix} 9 \\ 9 \\ 11 \end{pmatrix} + t \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} \text{ where } t \in \mathbb{R}.$$

(a) Show that lines L and M intersect at a point A and find the position vector of A .

```
s = symbols('s')
L = line(vec(1, 2, -3) + s*vec(2, 3, 6), s)
M = line(vec(9, 9, 11) + t*vec(4, 1, 2), t)

q6_3_pair = [...]      # [s, t] from two of the
                        # component equations
q6_3 = ...              # the position vector of
A

verify_pair('6.3 [s, t]', q6_3_pair, L, M)
verify_meet('6.3 A', q6_3, L, M)
```

6.4 — May 2021 TZ1 Paper 1 Q11(c), 7 marks

Consider the line L_1 defined by $\frac{x+1}{2} = y = 3 - z$ and the line L_2 defined by $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + t \begin{pmatrix} a \\ 1 \\ -1 \end{pmatrix}$, where $t \in \mathbb{R}$ and $a \in \mathbb{R}$.

It is given that the lines L_1 and L_2 have a unique point of intersection, A , when $a \neq k$.

(c) Find the value of k , and find the coordinates of the point A in terms of a .

```
a = symbols('a')
L1 = cartesian((x + 1)/2, y, 3 - z)
L2 = line(vec(0, 1, 2) + t*vec(a, 1, -1), t)

q6_4_k = ...      # k
q6_4_A = ...      # A in terms of a

verify_find('6.4 k', q6_4_k, a,
[no_unique_meet(L1, L2)])
verify_meet('6.4 A', q6_4_A, L1, L2)
```

§ 7. Parallel, meeting or skew

Four questions, 24 marks. Directions first; then two components and the third.

Answer the relation with one word: 'parallel', 'intersecting' or 'skew'.

7.1 — May 2021 TZ2 Paper 1 Q8(a), 3 marks

The lines l_1 and l_2 have the following vector equations where $\lambda, \mu \in \mathbb{R}$.

$$\begin{aligned} l_1 : \mathbf{r}_1 &= \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix}, & l_2 : \mathbf{r}_2 \\ &= \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \end{aligned}$$

(a) Show that l_1 and l_2 do not intersect.

Enter the reason they do not: the relation between the lines.

```
q7_1 = ...          # the relation between l1
and l2

verify_relation('7.1', q7_1, line(vec(3, 2,
-1) + lam*vec(2, -2, 2)), line(vec(2, 0, 4) +
mu*vec(1, -1, 1)))
```

7.2 — November 2025 TZ3 Paper 1

Q10(c), 5 marks

The point $P(-1, 1, -13)$ lies on the line L_1 ,

which has direction vector $\begin{pmatrix} 7 \\ 1 \\ 2 \end{pmatrix}$: $\mathbf{r} =$

$\begin{pmatrix} -1 \\ 1 \\ -13 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 1 \\ 2 \end{pmatrix}$. The line L_2 passes

through $A(2, -4, 2)$ and $B(7, -6, 1)$: $\mathbf{s} =$

$\begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ -2 \\ -1 \end{pmatrix}$.

(c) Show that L_1 and L_2 are skew.

Enter the relation, and $[\lambda, \mu]$ found from two of the component equations.

```

L1 = line(vec(-1, 1, -13) + lam*vec(7, 1, 2))
L2 = line(vec(2, -4, 2) + mu*vec(5, -2, -1))

q7_2 = ...           # the relation
q7_2_pair = [...]    # [lam, mu]

verify_relation('7.2', q7_2, L1, L2)
verify_pair('7.2 [λ, μ]', q7_2_pair, L1, L2)

```

7.3 — November 2022 Paper 2

Q12(c), 10 marks

Consider the points $A(1, 2, 3)$, $B(k, -2, 1)$ and $C(5, 0, 2)$, where $k \in \mathbb{R}$.

(c) For $k = 9$, let L_1 be the line passing through A , B and C .

(i) Find a vector equation of the line L_1 .

(ii) Line L_2 has the equation $\frac{x-1}{2} = \frac{y}{3} = 1-z$. Show that the lines L_1 and L_2 are skew.

For (ii) the parameter of L_2 is the common value μ of its three parts, and $[\lambda, \mu]$ comes from your line in (i).

```

A, C = vec(1, 2, 3), vec(5, 0, 2)
L2 = cartesian((x - 1)/2, y/3, 1 - z)

q7_3_line = ...      # L1: vec(...) +
lam*vec(...)
q7_3 = ...           # the relation
q7_3_pair = [...]    # [lam, mu]

verify_line('7.3(c)(i)', q7_3_line,
through(A, C))
verify_relation('7.3(c)(ii)', q7_3,
through(A, C), L2)
verify_pair('7.3(c)(ii) [ $\lambda$ ,  $\mu$ ]', q7_3_pair,
q7_3_line, L2)

```

7.4 — May 2025 TZ1 Paper 1 Q6, 6 marks

The line L_1 has vector equation $\mathbf{r} = 4\mathbf{i} - \mathbf{k} + \lambda(a\mathbf{j} + \mathbf{k})$, where $a, \lambda \in \mathbb{R}$.

The line L_2 has vector equation $\mathbf{r} = \mathbf{i} - b\mathbf{k} + \mu(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$, where $b, \mu \in \mathbb{R}$.

The lines L_1 and L_2 are perpendicular and intersect at a unique point.

Find the value of a and the value of b .

```
a, b = symbols('a b')
L1 = line(vec(4, 0, -1) + lam*vec(0, a, 1),
lam)
L2 = line(vec(1, 0, -b) + mu*vec(1, 2, 3),
mu)

q7_4 = [...]      # [a, b]

verify_find('7.4', q7_4, [a, b],
[perpendicular(L1, L2), meet(L1, L2)])
```

§ 8. Motion along a line

Two questions, 16 marks. $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$: speed, bearing, two times.

8.1 — May 2022 TZ2 Paper 2 Q11(a), (b), (d), 11 marks

Two airplanes, A and B , have position vectors with respect to an origin O given respectively by

$$\mathbf{r}_A = \begin{pmatrix} 19 \\ -1 \\ 1 \end{pmatrix} + t \begin{pmatrix} -6 \\ 2 \\ 4 \end{pmatrix}$$

$$\mathbf{r}_B = \begin{pmatrix} 1 \\ 0 \\ 12 \end{pmatrix} + t \begin{pmatrix} 4 \\ 2 \\ -2 \end{pmatrix}$$

where t represents the time in minutes and $0 \leq t \leq 2.5$. Entries in each column vector give the displacement east of O , the displacement north of O and the distance above sea level, all measured in kilometres.

(a) Find the three-figure bearing on which airplane B is travelling.

(b) Show that airplane A travels at a greater speed than airplane B .

The two airplanes' lines of flight cross at point P .

(d) (i) Find the coordinates of P . **(ii)** Determine the length of time between the first airplane arriving at P and the second airplane arriving at P .

In (a) enter the bearing as a string of three figures; for (b) both speeds.

```

rA = vec(19, -1, 1) + t*vec(-6, 2, 4)
rB = vec(1, 0, 12) + t*vec(4, 2, -2)
t1, t2, crossing = unknown('t1'),
unknown('t2'), unknown('crossing', 3)

q8_1a = ...      # the bearing of B
q8_1b_A = ...    # the speed of A
q8_1b_B = ...    # the speed of B
q8_1d_i = ...    # the point where the paths
cross
q8_1d_ii = ...   # the time between the
arrivals, minutes

verify_bearing('8.1(a)', q8_1a, velocity(rB))
verify_speed('8.1(b) A', q8_1b_A, rA)
verify_speed('8.1(b) B', q8_1b_B, rB)
verify_meet('8.1(d)(i)', q8_1d_i, line(rA,
t), line(rB, t))
verify_find('8.1(d)(ii)', q8_1d_ii, Abs(t1 -
t2), [Eq(rA.subs(t, t1), crossing),
Eq(rB.subs(t, t2), crossing)])

```

8.2 — May 2025 TZ1 Paper 2 Q7, 5 marks

At 09:00 a helicopter is located at a point $(10, 3, 0.5)$ relative to a point O on horizontal ground. The x -direction is due east, the y -direction is due north and the z -direction is vertically upwards. All distances are measured in kilometres.

The helicopter is flying at a constant height. Its position relative to O is given

by $\mathbf{r} = \begin{pmatrix} 10 \\ 3 \\ 0.5 \end{pmatrix} + 4t \begin{pmatrix} 10 \\ -25 \\ 0 \end{pmatrix},$ where t

represents the time in hours since 09:00.

(a) Find the speed of the helicopter.

At 10:00 the helicopter begins to descend. During descent the helicopter's vertical height decreases at a constant rate of 16 km h^{-1} and its horizontal velocity remains unchanged. The angle of descent, β , is defined as the angle between the helicopter's direction of travel and the horizontal.

(b) Find β , giving your answer in degrees.

```
r = vec(10, 3, 0.5) + 4*t*vec(10, -25, 0)
descending = velocity(r) + vec(0, 0, -16)
```

```
q8_2a = ...      # the speed, km/h
q8_2b = ...      # beta, degrees
```

```
verify_speed('8.2(a)', q8_2a, r)
verify_angle('8.2(b)', q8_2b, descending,
'horizontal', deg=True)
```



Solutions

Each answer is the one in the markscheme; the working is the shortest that earns the marks.

§ 1

1.1 (a) $\overrightarrow{BC} = (2, 4, 2) = \overrightarrow{AD}$: $D = (3, 0, 2)$.

(b) The midpoint of $[AC]$: $(0, -3, 2)$.

1.2 (a)(i) The midpoint of $[AB]$: $(-1, 3, -2)$. **(ii)** $AB = \sqrt{16 + 4 + 16} = 6$, and the radius is half of it: 3 .

1.3 $BV = \sqrt{9 + 16 + 81} = \sqrt{106} = 10.3$.

1.4 $VX = \sqrt{16 + 9 + 4} = \sqrt{29} = 5.39$.

1.5 (a) $\overrightarrow{AB} = (k - 1, -4, -2)$, $\overrightarrow{AC} = (4, -2, -1)$. **(b)** On one line, $\overrightarrow{AB} = 2\overrightarrow{AC}$: $k - 1 = 8$, $k = 9$.

1.6 (a) $|\mathbf{a}| = 13$: $2 \leq |\mathbf{a} + \mathbf{b}| \leq 28$. **(b)** $\mathbf{b} = -\frac{15}{13}\mathbf{a}$, so $\mathbf{p} = -\frac{2}{13}\mathbf{a} = \left(-\frac{24}{13}, \frac{10}{13}\right) = (-1.85, 0.769)$.

$$\mathbf{1.7} \quad \overrightarrow{AB} = \mathbf{c}; \quad \overrightarrow{OM} = \boxed{\mathbf{a} + k\mathbf{c}}, \quad \overrightarrow{MC} = \boxed{(1-k)\mathbf{c} - \mathbf{a}}.$$

§ 2

2.1 Perpendicular to $(12, -5)$ is $(5, 12)$, length 13; scaled to 15 with $x, y > 0$:

$$\boxed{\left(\frac{75}{13}, \frac{180}{13}\right)} = (5.77, 13.8).$$

2.2 $(\mathbf{a} + k\mathbf{c}) \cdot ((1-k)\mathbf{c} - \mathbf{a}) = (1-2k)\mathbf{a} \cdot \mathbf{c} - |\mathbf{a}|^2 + k(1-k)|\mathbf{c}|^2$, and with $|\mathbf{c}|^2 = 4|\mathbf{a}|^2$, $\mathbf{a} \cdot \mathbf{c} = 2|\mathbf{a}|^2 \cos \theta$:

$$\boxed{2(1-2k)|\mathbf{a}|^2 \cos \theta - |\mathbf{a}|^2 + 4k(1-k)|\mathbf{a}|^2} \\ = |\mathbf{a}|^2(1-2k)(2 \cos \theta - (1-2k))$$

$$\mathbf{2.3 (i)} \quad \boxed{-5p - 42} \quad \mathbf{(ii)} \quad \boxed{-8p - 54}$$

2.4 C is on the line OA and on F , and it is

$$\text{not } A: \quad \boxed{C\left(-kt, -\frac{k}{t}\right)}. \quad \text{Then } \overrightarrow{CA} =$$

$$\boxed{\left(2kt, \frac{2k}{t}\right)}, \quad \overrightarrow{CB} = \boxed{\left(kt - \frac{k}{t^3}, \frac{k}{t} - kt^3\right)}, \quad \text{and}$$

$$\overrightarrow{CA} \cdot \overrightarrow{CB} = 2k^2t^2 - \frac{2k^2}{t^2} + \frac{2k^2}{t^2} - 2k^2t^2 = 0$$

with $\overrightarrow{CB} \neq \mathbf{0}$ since $t \neq \pm 1$: $B\hat{C}A$ is a right angle.

§ 3

$$\mathbf{3.1} \quad \overrightarrow{VB} = (3, 4, -9), \quad \overrightarrow{VC} = (3, -4, -9):$$

$$\cos B\hat{V}C = \frac{74}{106}, \quad B\hat{V}C = \boxed{0.798} \quad (45.7^\circ).$$

$$\mathbf{3.2} \quad \mathbf{u} \cdot \mathbf{v} = \cos \frac{1}{n} + \sin \frac{1}{n}, \quad |\mathbf{u}| = \sqrt{2}, \quad |\mathbf{v}| = 1:$$

$$\boxed{\cos \theta = \frac{\cos \frac{1}{n} + \sin \frac{1}{n}}{\sqrt{2}}}.$$

$$\mathbf{3.3} \quad \mathbf{p} \cdot \mathbf{n} = 0: \boxed{\frac{\pi}{2}} \quad (90^\circ).$$

$$\mathbf{3.4} \quad \frac{-5p - 42}{\sqrt{74}} = \frac{-8p - 54}{\sqrt{145}}: \quad p = 4.78727 \dots =$$

$$\boxed{4.79}.$$

§ 4

$$\mathbf{4.1} \quad \textbf{(i)} \quad \frac{-1+1}{2} = 0 = 3 - 3. \quad \textbf{(ii)}$$

$$\boxed{\mathbf{r} = (-1, 0, 3) + \lambda(2, 1, -1)}$$

$$\mathbf{4.2} \quad \boxed{\mathbf{r} = (1, -2, 0) + \lambda(2, 3, 1)}$$

$$\mathbf{4.3} \quad (\mathbf{a}) \quad \boxed{\mathbf{r} = (-1, 1, -13) + \lambda(7, 1, 2)} \quad (\mathbf{b})$$

$$\boxed{\mathbf{s} = (2, -4, 2) + \mu(5, -2, -1)}$$

$$\mathbf{4.4} \quad -11 = 1 - 4\gamma, \quad 5 = 8 - \gamma, \quad 7 = -2 + 3\gamma:$$

$$\boxed{\gamma = 3}. \quad \text{With the direction } (4, 1, -3) \text{ it}$$

would be -3 .

§ 5

$$\mathbf{5.1} \quad \frac{|2a + 2|}{\sqrt{6}\sqrt{a^2 + 2}} = \frac{1}{\sqrt{2}} \text{ gives } a^2 + 8a - 2 = 0:$$

$$\boxed{a = -4 \pm 3\sqrt{2}}.$$

$$\mathbf{5.2} \quad \cos \theta = \frac{-28}{\sqrt{56}\sqrt{24}} = -0.7637\dots, \quad \theta = 139.8^\circ;$$

acute: $\boxed{40.2^\circ}$.

$$\mathbf{5.3} \quad \boxed{\cos \theta = \frac{2t}{\sqrt{2}\sqrt{(2t)^2 + (3+t)^2}}} = \frac{1}{2}, \quad \text{so}$$

$$4t = \sqrt{2}\sqrt{5t^2 + 6t + 9} = \sqrt{10t^2 + 12t + 18}.$$

§ 6

$$\mathbf{6.1} \quad -2 + 3\lambda = 4: \lambda = 2, t = 5: \boxed{(5, 4, 2)}.$$

$$\mathbf{6.2} \quad L_2: x = -1 + 2t, \quad y = 7 + t, \quad z = -5 + 3t;$$

$$5 + s = -1 + 2t, \quad 4 - s = 7 + t \quad \text{give} \quad t = 1,$$

$s = -4$, and $(5, 4, 2) - 4(1, -1, 1) = (1, 8, -2)$.

6.3 $1 + 2s = 9 + 4t$ and $2 + 3s = 9 + t$:

$s = 2, t = -1$; the third agrees, $9 = 9$.

$A(5, 8, 9)$.

6.4 $1 + 2t = at$ gives $t = \frac{1}{a-2}$: $k = 2$, and

$A\left(\frac{a}{a-2}, \frac{a-1}{a-2}, \frac{2a-5}{a-2}\right)$.

§ 7

7.1 $(2, -2, 2) = 2(1, -1, 1)$: parallel , and $(3, 2, -1)$ is not on l_2 — the first two components give $5 = 2$.

7.2 Not parallel; $\lambda = -1, \mu = -2$ from the first two, and $-15 \neq 4$: skew .

7.3 (i) $\mathbf{r} = (1, 2, 3) + \lambda(4, -2, -1)$ **(ii)**

$\lambda = \frac{1}{4}, \mu = \frac{1}{2}$, and $2.75 \neq 0.5$; the

directions are not multiples: skew .

7.4 $2a + 3 = 0$; then $\mu = 3, \lambda = -4$:

$a = -\frac{3}{2}, b = 14$.

§ 8

8.1 (a) $\boxed{063^\circ}$. **(b)** $\sqrt{56} = \boxed{7.48} > \sqrt{24} = \boxed{4.90}$ km/min. **(d)(i)** $t_1 = 2$, $t_2 = \frac{3}{2}$: $\boxed{P(7, 3, 9)}$. **(ii)** $\boxed{0.5}$ minutes.

8.2 (a) $4\sqrt{725} = 20\sqrt{29} = \boxed{108}$ km/h. **(b)** $\tan \beta = \frac{16}{20\sqrt{29}}$: $\boxed{\beta = 8.45^\circ}$.